

Letter to Millson

Dear Millson,

Nori brought me a copy of your paper with Goldman on deformations of flat bundles. It seems to me that it contains a number of inaccuracies, but also that, when corrected, it gives the stronger results you ask for on p. 8, lines -10 to -7. I begin with the latter.

The philosophy, which I had not realized before reading your paper, seems to be the following: in characteristic zero, a deformation problem is controlled by a differential graded Lie algebra, with quasi-isomorphic DG Lie algebras giving the same deformation theory. If the DG Lie algebra controlling a problem is “formal”, i.e. quasi-isomorphic to $\bigoplus H^i$, then the versal formal deformation space is that of $\bigoplus H^i$, i.e. the completion at 0 of the subscheme of H^1 defined by the equations $[u, u] = 0$.

If one believes this, what you do becomes transparent.

Deformation problem. G is a given reductive group, and one wants to deform a given G -torsor P . More precisely, one wants to deform a $G(\mathbb{C})$ -torsor, or equivalently a $G(\mathcal{O})$ -torsor, with integrable connection ∇ .

Controlling DG algebra. Over $\mathbb{C}$, the controlling DG algebra is

$$\Omega^{\bullet, \bullet}(\mathrm{Lie}(G^P)).$$

If P can be reduced to a maximal compact subgroup $K \subset G$, i.e. to a compact form of G , Hodge theory shows that the controlling algebra is formal: one has quasi-isomorphisms of DG Lie algebras

$$\Omega^{\bullet, \bullet}(\mathrm{Lie}(G^P)) \longleftarrow \mathrm{Ker}(d') \longrightarrow \mathrm{Ker}(d')/\mathrm{Im}(d'), \quad (1)$$

and $\mathrm{Ker}(d')/\mathrm{Im}(d')$ has $d = 0$.

To check the above, it is enough to use the principle of the two types, in the form that the double complex $\Omega^{\bullet, \bullet}(\mathrm{Lie}(G^P))$, forgetting the bracket, is a sum of

- a) a double complex with $d' = d'' = 0$ (the “harmonic part”);
- b) squares of isomorphisms

$$\begin{array}{ccc} * & \xrightarrow{d'} & * \\ d'' \uparrow & & \uparrow d'' \\ * & \xrightarrow{d'} & * \end{array} \quad (0 \text{ elsewhere}).$$

As far as formal deformations are concerned, this concludes the story. It is not clear to me how to use Hodge theory in the analytic case to get the same result for analytic-versal spaces. The algebraic case is all right, but Artin is more powerful than that.

Here is how to give a meaning to the philosophy, and how to check it.

Construction. Attach to a DG Lie algebra L^* over a field k of characteristic zero a category fibered over the category of spectra $\mathrm{Sp}(A)$, where A is an artinian local k -algebra of finite dimension with residue field k .

The objects over $\mathrm{Sp}(A)$, with $k = A/\mathfrak{m}$, are

$$\omega \in L^1 \otimes \mathfrak{m}, \quad d\omega + \frac{1}{2}[\omega, \omega] = 0.$$

Maps. The Lie algebra L^0 gives rise to a formal group $\widehat{L}^0$; the A -points of this formal group are $L^0 \otimes \mathfrak{m}$, viewed as a nilpotent Lie algebra and equipped with the Campbell-Hausdorff group law. Write this group as $\widehat{L}^0(A)$.

The formal group $\widehat{L}^0$ acts on L^1 through the bracket $[\cdot, \cdot] : L^0 \otimes L^1 \rightarrow L^1$, hence $\widehat{L}^0(A)$ acts on $L^1 \otimes A$. One also has a map

$$g^{-1}dg : \widehat{L}^0 \longrightarrow L^1, \quad \exp(\lambda) \longmapsto \frac{1 - \exp(-\mathrm{ad} \lambda)}{\mathrm{ad} \lambda} d\lambda,$$

and hence a map $\widehat{L}^0(A) \rightarrow L^1 \otimes \mathfrak{m}$.

One now defines a map from $\omega \in L^1 \otimes \mathfrak{m}$ to $\omega' \in L^1 \otimes \mathfrak{m}$ by taking $g \in \widehat{L}^0(A)$ and putting

$$\omega' = g dg^{-1} + g(\omega). \quad (2)$$

One has to check that

$$d\omega' + \frac{1}{2}[\omega', \omega'] = g \left(d\omega + \frac{1}{2}[\omega, \omega] \right)$$

when ω and ω' are related by (2).

To be checked. For a square-zero ideal I in A , and an object ω on A/I , there is an obstruction in $H^2(L \otimes I)$ to extending ω to A . If the obstruction vanishes, the isomorphism classes of extensions form a principal homogeneous space under $H^1(L \otimes I)$, and the automorphisms of any extension, trivial on A/I , are $H^0(L^* \otimes I)$. This is functorial in L^* , hence quasi-isomorphisms induce equivalences of fibered categories.

Application here. Let X be an analytic variety and let (V, ∇) be a holomorphic vector bundle with flat connection. The deformation problem is the functor which to A assigns the category of vector bundles with connection in the X direction, on $X \otimes \mathrm{Sp}(A)$, deforming (V, ∇) .

It is given by the preceding construction applied to

$$\Omega^{\bullet, \bullet}(\mathrm{End} V).$$

Remark. In your application, because you have the explicit quasi-isomorphism (1), you have an explicit equivalence of fibered categories between

- deformations of (V, ∇) as above;
- the category whose objects over $\mathrm{Sp}(A)$ are pointed maps

$$\mathrm{Sp}(A) \longrightarrow \{u \in H_L^1 \mid [u, u] = 0\},$$

and whose arrows are the elements $g \in \widehat{H}_L^0(A)$ carrying u to $g(u)$.

The functor upward is as follows. A cohomology class $u \in H^1 L \otimes \mathfrak{m}$, i.e. a map $u : \mathrm{Sp}(A) \rightarrow H^1 L$, can be written uniquely as an $\omega \in L^1 \otimes \mathfrak{m}$ with $d'\omega = d''\omega = 0$. This ω can be corrected by a $d'\rho$ to have zero curvature, if $[u, u] = 0$. The corrected $\tilde{\omega}$ is unique modulo $\exp(\mathrm{Ker} d' \otimes \mathfrak{m})$.

Remark. The quasi-isomorphism (1) is a filtered quasi-isomorphism for the Hodge filtration. This should translate into saying that the pair of spaces

$$(\text{deformations of } (V, \nabla)) \supset (\text{deformations of } V \text{ on a fixed } X)$$

is isomorphic to the pair

$$(H^1 \supset F^1 H^1) \cap \{[u, u] = 0\}.$$

II. Comments on your paper

p. 1, line 2 after “in B”: no. Take $G = U_1$.

p. 6, line 18, “Philosophic?”: I guess you want to start with η such that

$$d'\eta = d''\eta = 0.$$

p. 3, line 4: why does “closed” imply that the $(2, 0)$ -component is closed?

III. Generalization

Hodge theory is available for variations of Hodge structure, complex variations, on compact Kähler manifolds. The useful bigrading of $\Omega^{\bullet, \bullet}(\text{End } V)$ is obtained by mixing that of Ω and that of $\text{End } V$; it is useful now that (1) is a filtered quasi-isomorphism. One gets:

- a) the same result as before on deformations of a representation of π_1 given by a complex variation;
- b) looking at the deformation theory controlled by $F^0(\dots)$, the same result for the versal moduli of a filtered bundle with connection coming from a complex variation. One insists that the deformed connection continue to satisfy transversality:

$$\nabla F^i \subset \Omega^1 \otimes F^{i-1}.$$

Yours sincerely,

P. Deligne